Experiment 01

GPT Language Model Training Baseline Report

WikiText-103 subset | approximately 300K tokens | GPT-2 BPE / tiktoken | Google Colab T4

Report summary

This report documents the first controlled baseline for a from-scratch GPT-style decoder-only transformer. The experiment validated the training pipeline while revealing severe overfitting caused by limited data scale, high sample overlap, and incomplete checkpoint management.

Dataset	WikiText-103 subset
Token Count	~300K tokens
Tokenizer	GPT-2 BPE / tiktoken
Model Scale	~16M parameters
Epochs	5
Status	Baseline analysis and transition plan

1. Executive Summary

Experiment 01 established the first end-to-end baseline for the GPT-style language model training pipeline. The experiment was intentionally compact: it used a WikiText-103 subset of approximately 300K tokens, a GPT-2 BPE tokenizer, a context length of 128 tokens, and a four-layer decoder-only transformer with roughly 16 million parameters. The primary aim was implementation validation rather than final model quality.

The outcome was technically useful but scientifically limited. Training loss collapsed rapidly, confirming that the model and optimizer could fit the training data. At the same time, validation loss and perplexity increased sharply, demonstrating severe overfitting. This divergence made Experiment 01 a valuable failure-analysis baseline and directly informed the design of Experiment 02A.

~300K	~16M	0.2834	157,185
Tokens Small-scale corpus	**Model** Compact transformer	**Train loss** Memorization signal	**Val PPL** Severe overfitting

Principal finding: Experiment 01 validated the engineering pipeline but did not produce a generalizable language model. The training curve is best interpreted as evidence of memorization under data-limited conditions, not as evidence of language modeling competence.

2. Research Objective

The objective of Experiment 01 was to validate the minimum viable training stack for a GPT-style autoregressive transformer. The experiment tested whether the repository could correctly execute the sequence from raw text ingestion to tokenization, sample construction, causal self-attention, loss computation, optimization, validation, and checkpoint persistence.

The experiment specifically evaluated the following pipeline components:

Pipeline Component	Validation Goal
Dataset loading	Confirm that a text corpus can be loaded and split into training and validation streams.
GPT-2 BPE tokenization	Confirm compatibility with the tokenizer and vocabulary size expected by the model.
Sliding-window samples	Verify that fixed-length autoregressive training examples can be constructed.
Decoder-only transformer	Validate causal multi-head self-attention, residual blocks, normalization, and output logits.
Training loop	Confirm optimization, mixed precision, gradient clipping, and learning-rate scheduling.
Validation loop	Track validation loss and perplexity as the main generalization signals.
Checkpointing	Persist model state for later inspection and continuation.

3. Dataset and Experimental Context

The dataset was a small subset of WikiText-103 containing approximately 300K tokens. This dataset was suitable for fast iteration on Google Colab, but it was not large enough to support reliable generalization for a transformer with approximately 16 million parameters. The corpus was general-domain text, not specialized for any target downstream domain.

Field	Value
Dataset	WikiText-103 subset
Approximate size	Approximately 300K tokens

Field	Value
Approximate rows	5,000 rows
Domain	General English text
Tokenizer	GPT-2 BPE / tiktoken
Training platform	Google Colab
GPU	NVIDIA T4

The dataset size is the central limitation of Experiment 01. Small corpora can be useful for debugging, but they often cause transformers to memorize local patterns rather than learn robust distributions. This risk was amplified by the use of highly overlapped sliding windows, which increased the number of samples without proportionally increasing the diversity of unique language contexts.

4. Model Configuration

The model used a compact GPT-style decoder-only transformer. The architecture followed the standard autoregressive language modeling setup: token embeddings and positional embeddings are added, the sequence is processed through stacked causal transformer blocks, and final logits are produced for next-token prediction. The initial compact configuration used tied output weights and produced approximately 16 million parameters.

Component	Configuration	Approximate Parameters
Token embedding	50,257 x 256	12.9M
Position embedding	128 x 256	0.03M
Transformer stack	4 layers, d_model=256, heads=4	~3.1M
Language modeling head	Tied with token embedding	-
Total model	Decoder-only Transformer	~16M

Parameter	Value
Vocabulary size	50,257
Context length	128
Embedding dimension	256
Attention heads	4
Transformer layers	4
Dropout	0.1
Attention type	Standard causal self-attention
Flash attention	Not used
KV cache benchmark	Not performed

5. Training Configuration

The training setup used AdamW optimization, mixed precision, gradient clipping, gradient accumulation, and cosine learning-rate decay. These choices were sufficient for validating the mechanics of training, but the checkpointing policy was incomplete because only latest.pt was preserved. This meant that the best validation checkpoint could be overwritten by later epochs.

Hyperparameter	Value
Epochs	5
Optimizer	AdamW
Learning rate	3e-4
Minimum learning rate	1e-5
Weight decay	0.1
Gradient clipping	1.0
Gradient accumulation	4 steps
Mixed precision	Enabled
Learning-rate schedule	Cosine decay
Sliding-window strategy	Highly overlapped
Checkpoint strategy	latest.pt only

6. Training Results

Table 1 presents the observed training and validation metrics. The model fitted the training split very quickly, reducing training loss from 3.4662 to 0.2834. However, the validation loss increased monotonically from 8.6967 to 11.9652, and validation perplexity increased from 5,983 to 157,185.

Epoch	Train Loss	Validation Loss	Validation Perplexity
1	3.4662	8.6967	5,983
2	0.9319	10.7294	45,681
3	0.4473	11.5839	107,358
4	0.3232	11.8878	145,483
5	0.2834	11.9652	157,185

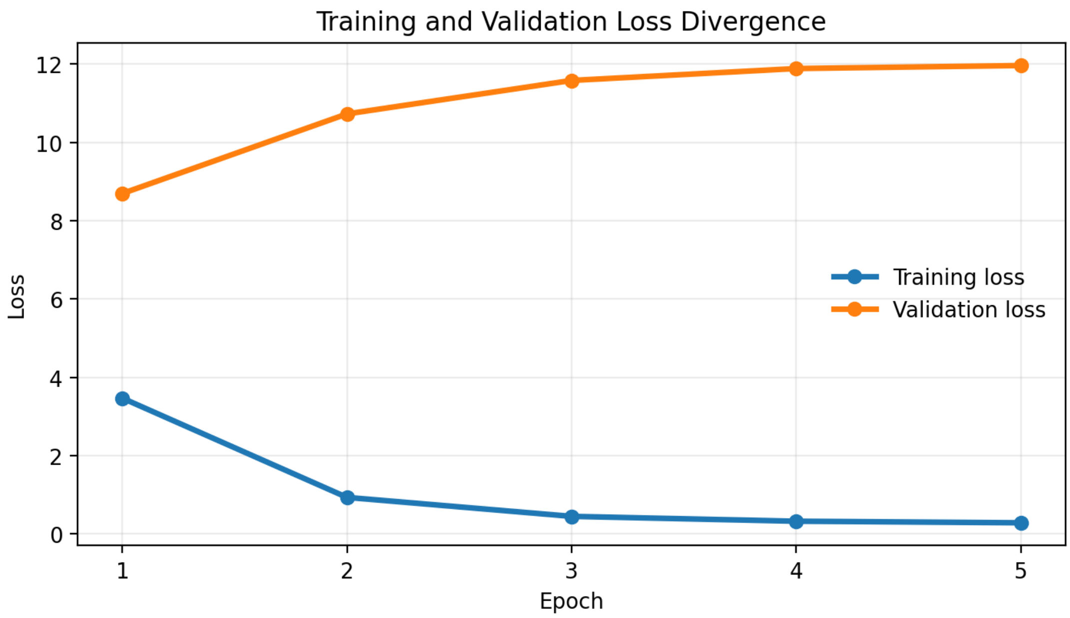

Figure 1. Training loss decreases while validation loss increases, indicating memorization rather than generalization.

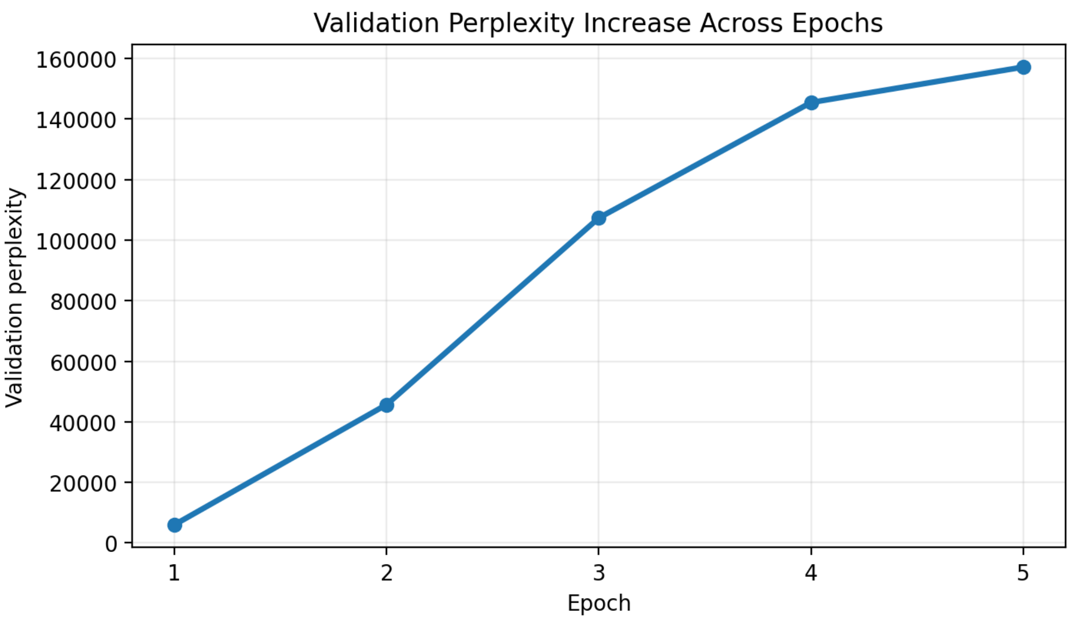

Figure 2. Validation perplexity increases sharply across epochs, confirming severe overfitting.

7. Training Behavior Analysis

The learning dynamics show a classic overfitting pattern. The model learned the training examples rapidly, which is visible from the steep decline in training loss. If training loss were considered alone, the experiment might appear successful. Validation metrics, however, show the opposite conclusion: the model became progressively worse on held-out data as training continued.

The increasing validation perplexity is particularly important. Perplexity is an exponential transformation of cross-entropy loss; therefore, even a moderate increase in validation loss can produce a very large increase in perplexity. By epoch 5, the validation perplexity exceeded 157,000, meaning the model assigned extremely low probability to the held-out validation tokens.

Interpretation: Experiment 01 should be considered a successful pipeline validation and an unsuccessful generalization experiment. The result is valuable precisely because it identified the failure mode early and made the redesign requirements clear.

8. Failure Diagnosis

The failure was not caused by a single issue. It resulted from an interaction between limited data scale, high sample overlap, insufficient checkpoint preservation, and the natural memorization capacity of transformer models. The following diagnosis summarizes the primary factors.

Issue	Severity	Technical Explanation
Dataset too small	High	Approximately 300K tokens was insufficient for reliable generalization at this model scale.
High sliding-window overlap	High	Overlapping windows increased the apparent number of samples but did not increase effective data diversity.
Context length too small	Medium	A 128-token context limited the model's ability to learn longer dependencies.
Checkpointing incomplete	High	Saving only latest.pt prevented systematic preservation of the best validation model.
No best checkpoint	High	The best-performing state could be overwritten by later overfitted states.
Limited evaluation scope	Medium	Loss and perplexity were tracked, but no inference-speed benchmark was available.
Colab runtime risk	Medium	Runtime interruptions created risk for long-running experiments and checkpoint retention.

9. Lessons Learned

Experiment 01 produced several engineering and research lessons that shaped the next stage of the project:

Lesson	Implication for Future Experiments
Training loss is insufficient.	Validation loss and perplexity must be treated as primary model-quality signals.
Small corpora encourage memorization.	The next experiment requires a substantially larger corpus.
Overlapping windows can be misleading.	Sample count should not be confused with true data diversity.
Checkpointing must be systematic.	best.pt, latest.pt, and epoch-level checkpoints should be preserved.
Context length matters.	The next baseline should use a longer context window.

Lesson	Implication for Future Experiments
Final experiments need reproducibility.	Logs, configs, checkpoints, and reports should be saved in structured experiment folders.

10. Redesign Requirements for Experiment 02A

Experiment 02A was planned as the direct successor to Experiment 01. Its purpose was to retain a simple tokenizer baseline while correcting the major limitations identified in the first experiment. The key question for Experiment 02A was whether the same training pipeline could generalize more effectively after improving data scale, context length, checkpointing, logging, and training infrastructure.

Area	Experiment 01 Limitation	Experiment 02A Improvement
Data scale	~300K tokens	Larger medical-domain corpus
Domain	General WikiText text	Medical-domain text
Context length	128 tokens	1024 tokens
Model scale	~16M parameters	Larger GPT-style baseline
Checkpointing	latest.pt only	best.pt, latest.pt, and epoch checkpoints
Logging	Basic metric tracking	Structured train and validation CSV logs
Evaluation	Loss and perplexity only	Add KV-cache inference benchmark
Platform	Colab T4	More stable GPU environment

Experiment 02A therefore becomes the first serious scaled baseline. It is not intended to replace Experiment 01; rather, it uses the lessons from Experiment 01 to establish a stronger and more reproducible medical-domain training setup.

11. Future Work: Experiment 02A

The immediate future work is Experiment 02A: a stronger baseline that keeps the GPT-2/tiktoken tokenizer while improving the experimental design. The focus is on testing whether the training pipeline generalizes better when the corpus is larger, the context window is longer, checkpointing is reliable, and evaluation includes both training metrics and inference behavior.

The central research question for Experiment 02A is:

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Can the same from-scratch GPT training pipeline achieve substantially better validation behavior
when dataset scale, context length, checkpointing, and infrastructure are improved?
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This transition keeps the scope clean: Experiment 01 remains the debugging and failure-analysis baseline, while Experiment 02A becomes the scaled baseline for the next stage of medical-domain GPT training.

12. Conclusion

Experiment 01 successfully validated the core GPT language modeling pipeline. It confirmed that the repository could load data, tokenize text, construct autoregressive samples, train a decoder-only transformer, compute validation metrics, and save checkpoints. From an engineering perspective, this made the experiment valuable.

From a model-quality perspective, the experiment failed to generalize. Training loss decreased substantially, but validation loss and perplexity increased dramatically. This divergence demonstrated that the model memorized the small training corpus rather than learning robust language representations.

The value of Experiment 01 lies in its diagnostic clarity. It exposed the exact weaknesses that needed to be corrected: data scale, context length, checkpointing policy, validation discipline, and experiment tracking. These findings directly motivated Experiment 02A, where the project moves to a larger medical-domain baseline with stronger reproducibility and more complete evaluation.

Appendix A. Repository Summary

Experiment 01 is best described in the repository as an initial baseline and failure-analysis study. The following summary can be used in README documentation:

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Experiment 01 validated the first GPT-style decoder-only transformer training pipeline on a small
WikiText-103 subset. The model reached a final training loss of 0.2834 but validation loss
increased to 11.9652 and validation perplexity increased to 157,185, demonstrating severe
overfitting. The experiment motivated the transition to Experiment 02A with a larger
medical-domain corpus, longer context length, improved checkpointing, structured logging, and
inference benchmarking.
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